

Shao-Shan (Seamus) Chang

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Education

University of Illinois Urbana-Champaign.

Champaign, IL

M.S. in Financial Engineering

Aug 2025 – Dec 2026

Coursework: Financial Computing, Statistical Methods, Machine Learning, Finance, Structured Credit, Financial Risk Management, Advanced Financial Derivatives, Stochastic Calculus

B.S. in Information Sciences + Data Science, GPA 3.93/4

Aug 2021 – May 2025

Dean's List: Fall 2021, 2022, 2024, Spring 2024; President of the UIUC Taiwanese Student Association

Coursework: Computer Science, Linear Algebra, Algorithms and Data Structures, Database Design and Prototyping

Skills

Programming: C++, SQL, Python, HTML/CSS, Javascript

Tools: Pandas, Numpy, Matplotlib, Scikit-learn, Django, Anaconda, Figma, GitHub

Experience

Consumer Default and Financial Ruin Modeling

Champaign, IL

Ernst & Young Practicum Project

Jan 2026 – Present

- Developing synthetic household-level financial data calibrated from BLS, Federal Reserve SHED, and BEA datasets to simulate monthly income, spending, and debt trajectories.
- Building ML models (Logistic Regression, Random Forest, XGBoost) with SHAP analysis to predict consumer financial distress over a 12–36 month horizon.
- Designing an optimization framework using binary search and simulation to compute personalized safe impulsive spending limits.

Compulsory Military Service – Taiwan Marine Corps

Pingtung, Taiwan

Ministry of National Defense R.O.C. (Taiwan)

Jun 2023 - Oct 2023

- Compulsory military service in the Marines Division of the Taiwan Navy Forces, developing strong discipline, adaptability, and teamwork skills through intensive training and operational deployments.
- Demonstrated professionalism, integrity, and work ethic while serving as part of the cohesive team, developed problem-solving and decision-making abilities in a high-pressure environment, and acquired a deep appreciation for national defense and the importance of leadership and dedication to mission objectives.

Landseed International Medical Group

Taoyuan, Taiwan

Software Development Engineer Intern

June 2022 – July 2022

- Learned ASP.NET, C#, MVC Design Pattern, Entity Framework and Web APIs through a 6-week intensive training.
- Developed a full-stack meeting room reservation system from scratch in 6 weeks, including features like searching/sorting data on table view, real-time database (SQL Server), interacting with business logic API, and third-party calendar plugin, etc.

Projects and Leadership

High Volatility Prediction Model for Crypto Market Making

Champaign, IL

IE 498 Machine Learning in Finance Course Project

Aug 2025 – Oct 2025

- Developed machine learning models, Logistic Regression, Random Forest, Gradient Boosting, to predict 1-second volatility spikes in ETH/USD market, achieving 89% accuracy and 86% precision from over 725,000 tick-by-tick observations.
- Engineered 11 market microstructure features including bid-ask spreads and order book imbalances, optimizing models by hyperparameter tuning to achieve ROC-AUC scores of 0.81-0.83, outperforming random baseline of 79% accuracy.

University of Illinois Urbana-Champaign Taiwanese Student Association

Champaign, IL

President

Jun 2022 – Jun 2023

- Coordinated with 5 department directors and over 40 members, while managing organizational sponsorships, financial operations, and event planning logistics.
- Partnered with global companies (TSMC, Marketech International Corp.) and over 5 local businesses (BakeLab, BrewLab, Latea, Golden Harbor, etc) to organize career programs, new student orientation, and cultural festival events.